PVP 14

Code: CS1T3, CE1T3, IT1T3, EC1T4

I B.Tech - I Semester – Regular Examinations – November 2015

ENGINEERING CHEMISTRY

(Common for CSE, CE, IT, ECE)

Duration: 3 hours Max. Marks: 70

PART – A

Answer ***all*** the questions. All questions carry equal marks

11x 2 = 22 M

1. a) Distinguish between temporary and permanent hardness of Water.
   b) Write short note on ion exchange process.
   c) Express the term break-point Chlorination.
   d) Write any four engineering applications of polymers.
   e) Explain disproportianation in addition polymerization.
   f) Distinguish between thermoplastic and thermosetting resins.
   g) What is the need of green chemistry?
   h) What is paint? What are the good characteristics of a good paint?
   i) Write any three factors that influence corrosion with respect to the nature of Environment.
   j) Define solar energy. Write two applications.
   k) What are high temperature super conductors?

PART – B

Answer any ***THREE*** questions. All questions carry equal marks. 3 x 16 = 48 M

2. a) Write any four differences between cold lime soda and hot lime soda process. Calculate the amount of lime and soda required for the treatment of 10,000 litres of raw water containing the following dissolved salts per litre $Ca(HCO_3)_2$ =16.2 mg, $CaCl_2$ = 11.1 mg, $MgSO_4$ = 12 mg, $NaHCO_3$ = 7.25 mg, CO_2 = 4.4 mgs. 8 M

b) What is meant by sterilization of water? Explain how sterilization of water Carried out by using bleaching powder and ozone? 8 M

3. a) Describe the method of preparation and properties of PVC, Bakelite and Polycarbonates. 8 M

b) Write the preparation, properties and engineering applications of any one Bullet proof plastic. 8 M

4. a) What is green chemistry? Explain aqueous phase green synthesis with suitable example. 8 M

b) What are fullerenes? Explain Carbon Bucky balls. Write any five applications of nanomaterials. 8 M

5. a) What is corrosion? Describe galvanic and differential aeration corrosion. 8 M

b) Explain the electrochemical theory of wet corrosion giving its mechanism. 8 M

6. a) Write different types of liquid crystals. Discuss their engineering applications. 8 M

b) What are the different types of semiconductors? Explain the controlled valance Semiconductor and organic semiconductor. 8 M

Code: CE1T3, CS1T3, EC1T4, IT1T3

I B. Tech - I Semester – Regular/Supplementary Examinations
December 2016

ENGINEERING CHEMISTRY
(Common for CE, CSE, ECE & IT)

Duration: 3 hours Max. Marks: 70

PART – A

Answer ***all*** the questions. All questions carry equal marks
11 x 2 = 22 M

1.
a) State the impurities that cause hardness of water.
b) Why is hot lime-soda process better than the cold process?
c) Give any two examples of biodegradable polymers.
d) Mention the moulding constituents of plastics.
e) What is sonochemistry? Explain.
f) Explain any two engineering applications of fullerenes.
g) Write a short note on dry corrosion.
h) What are the characteristics of a good paint?
i) Discuss briefly on Superconductors.
j) Give any four applications of liquid crystals.
k) What is Doping? Explain with one example.

PART – B

Answer any ***THREE*** questions. All questions carry equal marks.
3 x 16 = 48 M

2. a) Calculate the temporary and permanent hardness water containing $Mg(HCO_3)_2$=29.2mg/L, $Ca(HCO_3)_2$= 32.4mg/L, $CaCl_2$ =27.75mg/L, $MgSO_4$=24mg/L, KCl=20mg/L. 6 M

b) Distinguish between hot lime-soda and cold lime-soda process. 4 M

c) Illustrate the reverse osmosis process with a diagram. 6 M

3. a) Enumerate the differences between addition and condensation polymerization. 4 M

b) Write a brief note on the preparation and properties of Bakelite. 6 M

c) Describe any two moulding techniques for fabrication of plastics. 6 M

4. a) Write the principles of green chemistry. 8 M

b) What are carbon nanotubes? Explain their types. 8 M

5. a) Discuss the mechanism of electrochemical corrosion. 4 M

b) Write notes on

i) Differential aeration 4 M

ii) Sacrificial anodic protection 4 M

iii) Galvanizing 4 M

6. a) Explain the stoichiometric and non-stoichiometric defects in crystals. 8 M

b) Write a detailed note on liquid crystal display. 4 M

c) Write a brief account on photo voltaic cells. 4 M

PVP 14

Code: CE1T3, CS1T3, EC1T4, IT1T3

I B. Tech - I Semester – Regular/Supplementary Examinations
November 2017

ENGINEERING CHEMISTRY
(Common for CE, CSE, ECE & IT)

Duration: 3 hours Max. Marks: 70

PART – A

Answer ***all*** the questions. All questions carry equal marks

11 x 2 = 22 M

1.
a) Define hardness of water? What are the causes of hardness?
b) Which one is most widely used chlorination process?
c) How the plastics are classified?
d) Write the name and structure of bullet proof plastic.
e) Give the applications of nonmaterial in food industries.
f) Write few characteristics of fullerenes.
g) What are the main functions of the paint?
h) Steel screw in a brass marine hardware corrodes. Explain.
i) Give the importance of pilling bed worth rule.
j) What are liquid crystals?
k) What are high TC super conductors?

PART – B

Answer any ***THREE*** questions. All questions carry equal marks.

3 x 16 = 48 M

2.a) One liter of water from an underground reservoir in Tirupathi Town in Andhra Pradesh showed the following analysis for its contents: $Mg(HCO_3)_2$ = 42 mg; $Ca(HCO_3)_2$ = 146mg; $CaCl_2$= 71mg; NaOH = 40mg; $MgSO_4$ = 48 mg; organic impurities = 100 mg; Calculate temporary, permanent and total hardness of this sample of 10,000 liters of water. 8 M

b) How is the exhausted resin regenerated in an ion exchanger? What are the merits and demerits of ion exchange method? 8 **M**

3.a) Mention the compounding materials used in plastics indicating their function and give one example for each**?** 8 **M**

b) Discuss the preparation, properties and applications of Bakelite. 8 **M**

4.a) What is green Chemistry? What are the twelve principles of green chemistry? 8 **M**

b**)** Green Chemistry is sustainable chemistry- Explain the statement**.** **4 M**

c) What do you mean by nano materials? Explain different categories of nano materials with suitable examples. **4 M**

5.a) Define pitting corrosion. State the conditions necessary for pitting corrosion to occur. State the hazards and controls associated with pitting corrosion. 8 **M**

b) What do you mean by sacrificial anodic protection? Explain with suitable examples. 8 **M**

6.a)What are liquid crystals? What are the advantages of liquid crystal display? **5 M**

b) Discuss the effect of temperature on semi conductors. 3 M

c) What do you understand by a semi conductor? Discuss some properties of semi conductors**.** 8 **M**

PVP 14

Code: CE1T3, CS1T3, EC1T4, IT1T3

I B. Tech - I Semester – Regular/Supplementary Examinations
November 2018

ENGINEERING CHEMISTRY
(Common for CE, CSE, ECE & IT)

Duration: 3 hours Max. Marks: 70

PART – A

Answer ***all*** the questions. All questions carry equal marks
11 x 2 = 22 M

1.
a) Distinguish between Carbonated Non Carbonate hardness of water.
b) Why di-sodium salt of EDT A is used instead of EDTA in the estimation of hardness of water in EDTA method and also why the buffer is used?
c) Write the structure of Bakelite.
d) Write about the significance of Economy in the principle of green chemistry.
e) Write the structure of C60 Fullerene.
f) Write the significance of Pilling-Bedworth rule.
g) Write the applications of galvanization and tinning.
h) What is the advantage of super conductor give one example.
i) What is greenhouse effect? Name the greenhouse gases.
j) Write two applications of LCD
k) Write two properties of Nano materials.

PART – B

Answer any ***THREE*** questions. All questions carry equal marks.

3 x 16 = 48 M

2. How is hardness of water removed by
   a) Ion Exchange process
   b) Zeolite process and write the reactions involved ?

16 M

3. a)Write the mechanism of Ionic Polymerization. 8 M

   b) Write the preparation, properties and uses of Bakelite

8 M

4. Write briefly
   a) Super critical fluid extraction method and
   b) Micro wave induced method in detail with figures.

16 M

5. a) What is corrosion? Describe galvanic and differential aeration corrosion with diagrams and examples. 8 M

   b) Explain sacrificialanodic protection and impressedcurrent cathodic protection. With diagrams and advantages. 8 M

6. a) Write definition of Semiconductors and explain types of Semiconductors. 8 M

b) Explain greenhouse concepts in detail. 8 M